

Jackson Barber

j7barber@uwaterloo.ca

Toronto, Ontario. (416) 471-9656. [Linkedin](#) [Github](#)

Education

University of Waterloo - BAsC in Mechatronics Engineering - 3.92 GPA Sept 2025 – Sept 2030

- Awards: David Gerard Conroy Undergraduate Engineering Scholarship

Skills

- **Modeling / Simulation:** Solidworks, Fusion360, OnShape, Autocad, 3D Printing, Traditional & Master Modeling, Ansys
- **Manufacturing:** CNC Mill, Manual Mill, Lathe, Thermoformer, Bandsaw, Mig & Arc Welding, Carbon Fibre Layup
- **Electrical / Controls:** Kicad, Altium, Soldering, Crimping, Sensor Use, Stepper and Servo Motor Use
- **Robotics & Vision:** FANUC Roboguide, Studio 5000, VisionMaster, VSCreator, SigmaWin

Experience

Mechatronics Engineering Co-op – Precision Resource - Cambridge, ON May 2026 - August 2026

- Designed and assembled a universal robotic inspection workcell including a frame, tooling, camera mounts, and conveyor integration allowing for photo inspection of new parts.
- Trained an image segmentation AI model to detect defects in a high volume production part with a 99.5% accuracy over a 2000 part test with 0 false positives
- Utilized FANUC Roboguide and IrProgrammer to simulate, program and integrate an SR-3ia, cutting cycle time by 25%
- Modelled and assembled a custom frame for camera and robot mounting, reduced vibrations allowed for 24% more accurate repeatability

Mechanical Subteam Lead – Waterloo Aerial Robotics Group (WARG) - Waterloo, ON Sept 2025 - Present

- Led the design of the team's composite fixed-wing aircraft by creating master models, design tables, and detailed design of the empennage.
- Led the manufacturing of 5+ CF layups including foam core and split mold airfoils, as well as sandwich panels
- Designed and optimized an avionics lid for the competition drone using surface modeling and draft analysis in SolidWorks, reducing weight by 59% while maintaining structural integrity
- Developed and simulated a custom firefighting nozzle using CFD analysis, increasing exit velocity by 53%
- Designed a CNC hotwire cutter for custom wing designs, 4 axis construction allowed for taper and sweep not possible with manual methods

Lead Designer – WARP7 Robotics Team - Toronto, ON Sept 2022 - May 2025

- Led strategic design by developing pros-cons analysis templates, match strategy, and data-driven scouting documents
- 3D modeled and manufactured manipulator mechanisms in SolidWorks, drafting technical drawings with GD&T specifications and fabricating components using CNC and manual machining processes
- Developed educational resources and ran technical seminars, achieving a 33% increase in rookie member engagement
- Implemented skeleton sketches and master modeling workflows to enable parametric design iteration

Projects

Custom Smart Watch

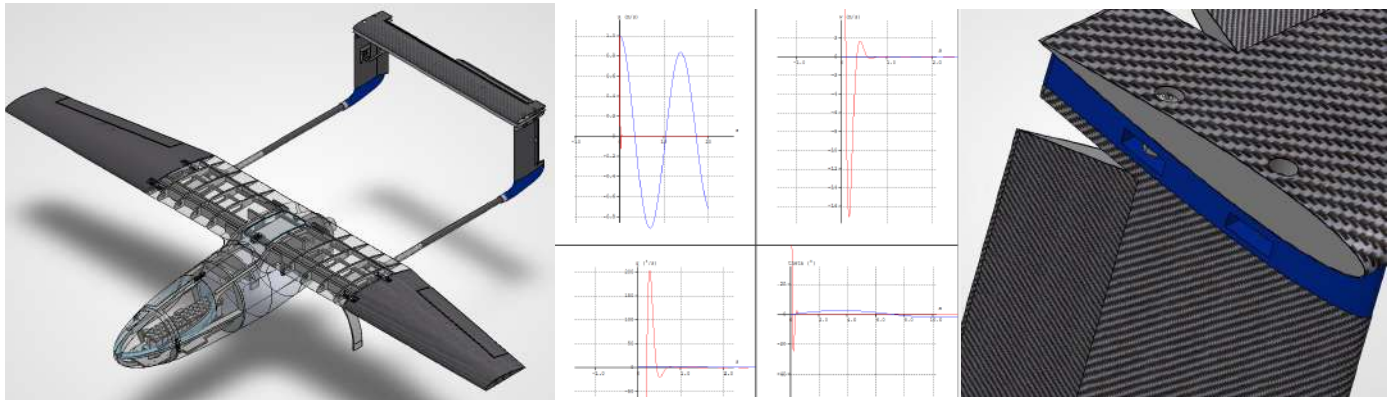
- Designed a custom smart watch PCBA & case allowing for full control over firmware, controllability, and power use
- Use of the nRF54L15 MCU allows for 700% lower active current draw compared to fitbit, use of low power sensors allows for a total battery life increase of 130%/mah
- Features include: step counting, music control, touch screen, heart rate sensing, USB-C charging, and notification vibration

Precision Disc Launcher

- Designed and manufactured a precision disc launcher with dual-axis spin control (pitch and yaw), achieving 5cm accuracy at 10m target distance for competitive applications
- Implemented a novel outrunner motor configuration with the motor body directly spinning the launching wheel, reducing overall assembly size and weight by 30% compared to traditional shaft-driven designs

Helios M2 Empennage Design @ WARG

- Sized an Inverted U-Tail Hstab, Vstab, and control surfaces using Sadreay and Caughey's methods ensuring stability and controllability even at critical conditions
- Completed lift and stability analysis at cruise and stall conditions in XFLR5 ensuring stable natural response for all four modes
- Completed hinge moment calculations to inform servo sizing for control surfaces at different load conditions
- Utilized Solidworks Excel design tables to allow for data to automatically update as input values changed using a custom toolchain
- Parameterized and created an XML file defining flight dynamics to allow for SITL flights in JSBSIM



Automated Part Inspection @ Precision Resource

- Developed a custom Automated Visual Inspection system allowing for automatic pass/fail judgement of parts
- Achieved an accuracy rate of 99.5% over a test run of 2000 parts with 0 false positives
- Projected to be implemented on a production line of ~1,000,000 parts/qtr
- Trained and implemented an AI model for detecting defects increasing accuracy from 96% -> 99.5% over rule based systems
- Designed and built a custom frame allowing for seamless integration of all required components and sensors
- Implemented beam brakes and air knives for automatic photo triggers and automated part rejection
- Logged all inspection results using an FTP server to allow for detailed visualization of failure modes common to each production run



*Marketing image used as to not reveal part

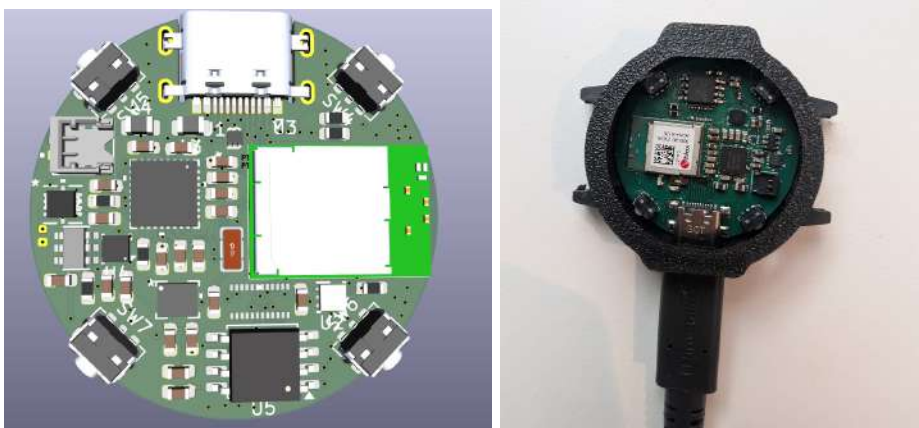
Automated Part Sorting @ Precision Resource

- Developed a custom universal sorting system, allowing any parts placed in a hopper to be serialized, and placed by a robot
- Implementation of automated system projected to save ~\$300,000/yr in operator costs
- Designed and assembled custom tooling allowing for maximum compatibility, improved design reduced tool changes by 33%
- Modelled and assembled a custom frame for camera and robot mounting, reduced vibrations allowed for 24% more accurate repeatability
- Wrote TP programs to allow the FANUC robot to locate, pick, and place parts
- Utilized ROBOGUIDE to simulate potential robot pathing ensuring no collisions on physical robot
- Began PLC integration to allow robot to call a Flexibowl and gooseneck conveyor to refill parts



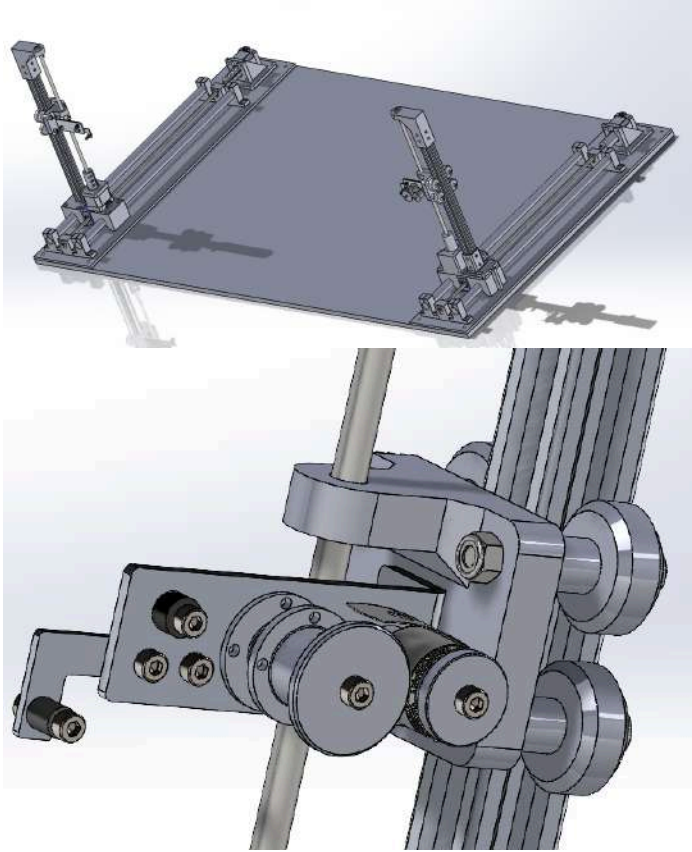
Custom Smart Watch

- Designed a custom smart watch to allow for more control over software and hardware integration
- Incorporates step counting, music control, touch screen, heart rate sensing, USB-C charging, and notification vibration
- Built around the nRF54L15 to allow for bluetooth connectivity while staying at low power
- Contained in a custom printed case allowing for a 30% lighter design and personal customization
- 250mah battery allows for a projected life of 24hrs
- Built off of Zephyr RTOS to allow for real time sensor communication and bluetooth connection



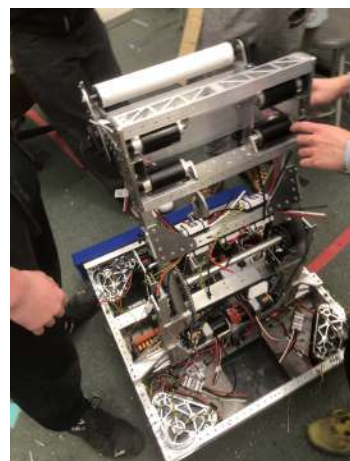
CNC Hotwire Cutter @ WARG

- Designed a CNC hotwire cutter to allow for easier wing design for fixed wing aircraft
- Incorporated an anti-backlash nut using a set screw to prevent slop
- Designed an adjustable limit switch to ensure an accurate homing sequence
- Run through an SKR Pico using a modified version of the GRBL CNC library
- Adapted custom DXF airfoil profiles into Gcode using JediCut



FRC 2024 Disc Launcher @ WARP7

- Personally designed and manufactured a manipulator to shoot foam rings capable of applying spin in both the pitch and yaw axis.
- Used a unique outrunner design having the wheel spin on the motor rather than through a shaft. This design allowed for a 30% reduction in size and 25% reduction in weight when compared to the average team.
- Integrated a hall effect sensor to allow for autonomous shooting sequence
- Designed using type 2 interference fits to ensure strong hold while not damaging the attached motor
- Fabricated the parts using both CNC and manual milling machines



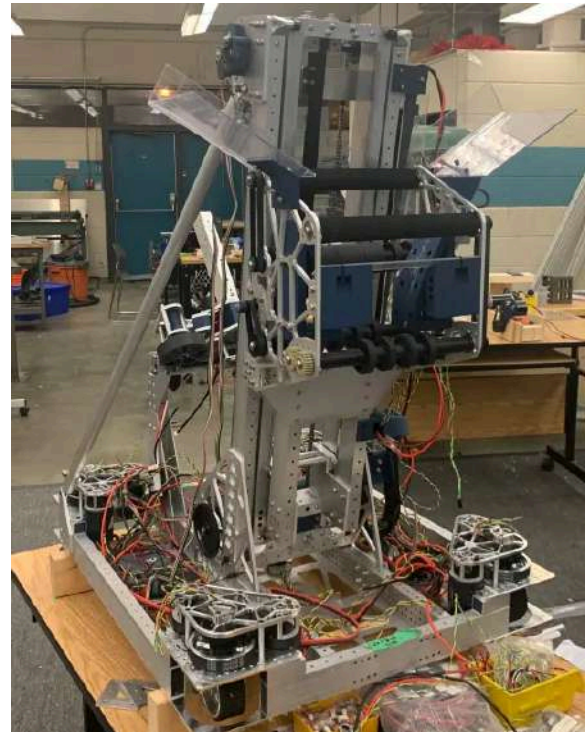
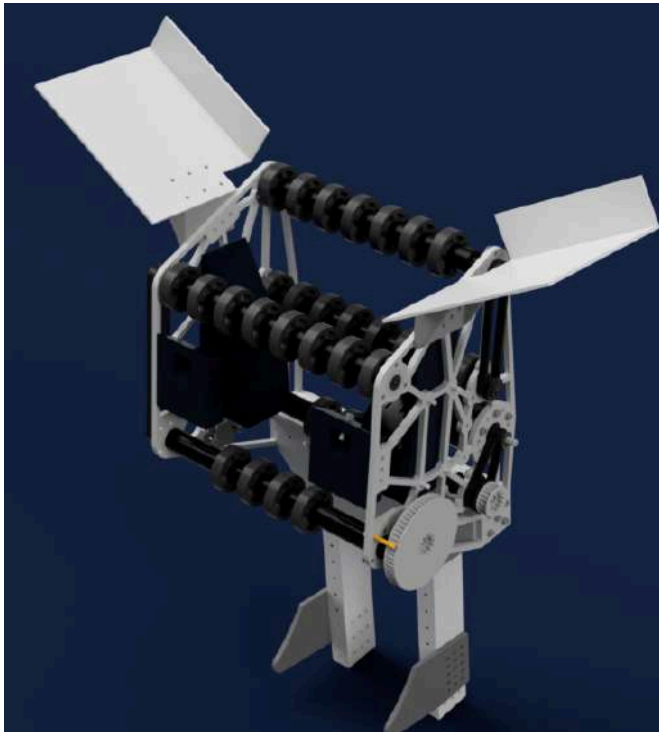
Custom Machined Aluminium Chess Set

- Manually machined a custom aluminium chess set
- Worked off engineering drawings to ensure piece consistency
- Measured pieces using micrometers, calipers, and machine DRO



FRC 2025 PVC Pipe Manipulator @ WARP7

- Personally designed and fabricated the intake/outtake mechanism using a master modelling workflow
- Designed modularly to allow for additional DOF, change of motors, or increased compression
- Designed polycarbonate and 3DP guides to allow for intake at any angle -> eject vertically downward
- Incorporated a second front intake that allowed it to hold and shoot yoga balls into a net
- Created custom pocketing pattern reducing plate weight by 68% while maintaining structural stability



Avionics Lid @ WARG

- Developed the waterproofing lid for the 2026 competition drone using solidworks surfaces and draft analysis
- Manufactured by thermoforming 0.020" PETG on a custom routed XPS mold
- Included a flexure near the mounting point to reduce load and prevent snapping
- Custom clip mounting system allowed for easy mounting
- Added relief holes to ensure strong suction while thermoforming
- Flew at AEAC 2026, ensuring reliability under rainy conditions

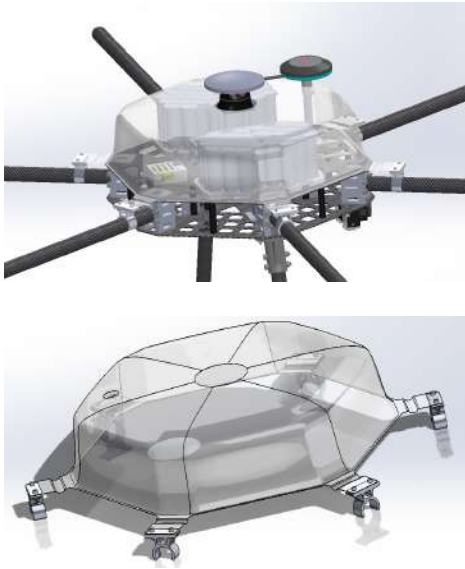


Table Tennis Serving Robot

- Designed a robot capable of serving ping pong balls for serve receive practice
- Runs from a single USB-C connection and is controlled using a custom mobile app
- Designed boards for: USB-C power negotiation, motor and microcontroller breakout, and buck voltage regulation
- Uses a printed spindler and wheels to allow for spin control
- Mobile app was designed using Expo for React Native allowing for simple end user bluetooth connection
- Fully modular electronic design allows for reusability of power boards

